

Summarizing Distributions with Center, Spread and Shape

Center, spread, shape, and reading a display

Directions:

- Both data sets below record daily reading time, in minutes, already sorted from least to greatest.
- Write each answer on its answer line, with the unit.
- Where a problem asks you to describe a distribution, name the center, the spread, and the shape.

Class A (10 students, minutes)	20, 25, 30, 30, 35, 40, 40, 45, 50, 85
Class B (7 students, minutes)	26, 30, 34, 38, 42, 46, 50

1. Find the mean daily reading time for Class A.

Answer: _____ minutes

2. Find the median daily reading time for Class A.

Answer: _____ minutes

3. Find the range of the Class A times.

Answer: _____ minutes

4. A school library recorded the books borrowed each day one week:

Day	Mon	Tue	Wed	Thu	Fri
Books borrowed	12	19	15	22	27

- (a) Find the total number of books borrowed that week. _____
- (b) Find the number borrowed on the busiest day. _____

5. For Class A, find the quartiles and the interquartile range.

- (a) The first quartile Q_1 .

Answer: _____ minutes

- (b) The third quartile Q_3 .

Answer: _____ minutes

- (c) The interquartile range $Q_3 - Q_1$.

Answer: _____ minutes

6. A value is called an outlier when it lies more than $1.5 \times \text{IQR}$ beyond a quartile.

- (a) Compute the upper fence $Q_3 + 1.5 \times \text{IQR}$ for Class A.

Answer: _____ minutes

- (b) Name the Class A value that lies beyond that fence, and describe the shape of the Class A distribution.

7. Now work with Class B.

(a) Find the mean.

Answer: _____ minutes

(b) Find the median.

Answer: _____ minutes

(c) Explain what the agreement of these two numbers says about the shape of Class B, and how that differs from Class A.

8. Find the standard deviation of the Class B times, treating the seven values as the whole population. Round to the nearest whole minute.

Answer: _____ minutes

9. One of the ten Class A students is chosen at random. Find the probability that the chosen student read more than 45 minutes.

Answer: _____

10. Synthesis. The 85 in Class A turns out to be a recording error: the student actually read 45 minutes. The corrected list is 20, 25, 30, 30, 35, 40, 40, 45, 45, 50.

- (a) Find the new mean.

Answer: _____ minutes

- (b) Find the new median.

Answer: _____ minutes

- (c) Explain why one of these two measures moved a great deal and the other did not move at all.